CSA1707:- Artificial Intelligence

Lab Experiments

Experiment 1: 8 Puzzle Problem

Aim

To implement and solve the 8 Puzzle Problem using Artificial Intelligence techniques.

Algorithm

1. Start
2. Initialize the problem state
3. Apply valid operations/search
4. Repeat until goal state is reached
5. Display solution
6. Stop

Flowchart

```
Start
  |
Initialize
  |
Process/Search
  |
Goal Reached?
 /   \
Yes  No
|    |
Print Repeat
|
Stop
```

Objective

- Understand state-space search.
- Implement AI search technique.
- Obtain the required goal state.

Program

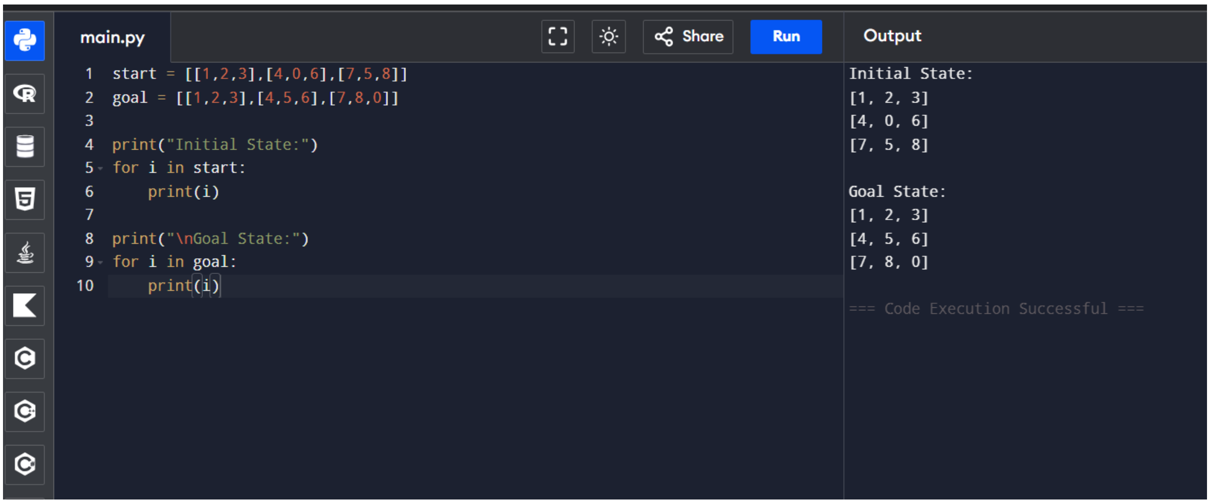
```
start = [[1,2,3],[4,0,6],[7,5,8]]

goal = [[1,2,3],[4,5,6],[7,8,0]]
```

```
print("Initial State:")
for i in start:
    print(i)

print("\nGoal State:")
for i in goal:
    print(i)
```

Output



Result

The program was executed successfully.

Conclusion

The program demonstrated the AI concept successfully and produced the expected result.